

CCNA 200-301 V2.0

01 Networks, Models and This Exam

Part I — Foundations

LECTURER

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What you will be able to do

- **Describe** what a network does, in terms of the four things every network must solve: naming, addressing, forwarding and error handling.
- **Explain** the layered model well enough to use it as a troubleshooting tool, which is the only reason this exam still cares about it.
- **Identify** the devices in a small enterprise network and say what each one decides.
- **Navigate** the CCNA v2.0 blueprint and this book against it.

Before we start

Nothing. This is the first chapter, and the course assumes no prior networking.

Vocabulary for this chapter

host

node

link

frame

packet

segment

encapsulation

layer

protocol

switch

router

access layer

distribution layer

broadcast domain

collision domain

Each is defined where it first matters — not here.

The four problems, and the four answers

Every network technology in this book is an answer to one of four questions.

- **Who am I talking to?** → addressing. MAC addresses locally (Chapter 3), IP addresses globally (Chapter 4).
- **How do I find them?** → forwarding. Switches learn (Chapter 3); routers decide (Chapter 15).
- **How do we not talk over each other?** → media access and loop prevention (Chapter 3, Chapter 12).
- **What if it goes wrong?** → error detection, acknowledgement and retransmission (Chapter 6).

When you meet a new protocol, ask which of the four it is answering. It will be one of them.

Protocol

An agreed set of rules for a conversation: what may be said, in what order, in what format, and what to do when the expected reply does not arrive.

Two people who both know to say “hello”, wait, then speak, have a protocol. Two computers need theirs written down to the bit, because neither can improvise. Almost every capitalised acronym in this book — ARP, DHCP, OSPF, HSRP — is one such agreement, and each exists because somebody had one of the four problems above.

The model is a troubleshooting tool

When something does not work, the layered model tells you *what to test next*. If a cable is unplugged, checking DNS is a waste of a minute you do not have in the exam. Work up: is the link up? Does it have an address? Can it reach its gateway? Can it resolve a name? Every troubleshooting question in this exam — and there are eight topics' worth — is answered by walking the layers in order.

The v2.0 blueprint no longer asks you to *describe* the OSI model, as v1.1 did. It assumes you use it.

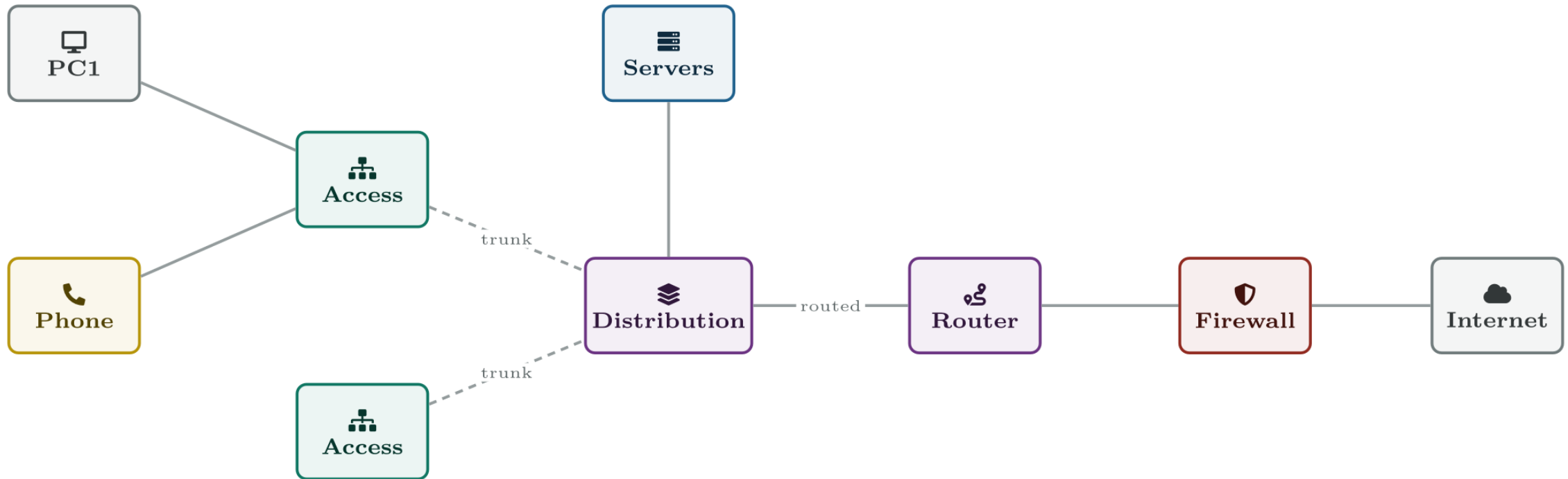
Encapsulation, and the names of things

As data moves down the stack each layer wraps it in a header. The wrapped unit gets a different name at each layer, and using the right one is how you show you know which layer you are discussing.

- Application data becomes a **segment** at layer 4 (TCP) or a **datagram** (UDP), when a port number is attached.
- That becomes a **packet** at layer 3, when IP addresses are attached.
- That becomes a **frame** at layer 2, when MAC addresses are attached.
- That becomes **bits** at layer 1.

Switches forward frames. Routers forward packets. That single sentence explains most of the difference between the two devices.

The devices, and what each one decides



A small enterprise network. Access switches connect end devices; a distribution switch aggregates them and routes between VLANs; the router reaches other networks; the firewall polices what leaves. Every chapter in Parts II to IV configures one piece of this picture.

Two domains worth keeping straight

A **collision domain** is the set of ports that could transmit at the same instant and interfere. Every switch port is its own collision domain, which is why modern switched networks do not collide — and why a **collisions** counter that is climbing means something is badly wrong, usually a duplex mismatch (Chapter 2).

A **broadcast domain** is the set of devices that receive each other's broadcasts. One VLAN is one broadcast domain. Splitting broadcast domains is the entire purpose of Chapter 8, and joining them again is the purpose of Chapter 15.

How the blueprint is built, and how to read it

This is what the blueprint actually asks for

Cisco publishes twenty-nine numbered topics in five domains. The number is the whole contract: if a technique is not in a topic statement, it is not directly examinable, and if it is, the *verb* tells you the depth expected.

- *Describe* — you must be able to explain it. Six topics.
- *Interpret, Validate, Select, Use, Manage* — you must be able to read output or make a choice. Seven topics.
- *Configure* — you must produce the commands. Eight topics.
- *Troubleshoot* or *Diagnose* — you must find and fix a fault. Eight topics.

Sixteen of twenty-nine are configure-or-fix. That is the single most useful fact about this exam, and it should change how you study: time at a command line beats time with a highlighter, by a wide margin.

This exam, and this book

- **Studying a v1.1 book.** The exam number, 200-301, did not change when the blueprint did. A book that teaches QoS, NTP, REST APIs, JSON or Terraform in a CCNA context is written for the old blueprint. Check the version before you trust the contents.
- **Learning the OSI layers as a list to recite.** Nobody will ask you to name layer 6. They will show you a broken network and expect you to work upwards.
- **Assuming “switch” means layer 2.** Multilayer switches route. Most of the distribution layer in a real network is switches doing routing, and topic [2.1](#) explicitly includes switch virtual interfaces.

Three reports, three different layers, and no commands yet.

You have not configured anything and cannot yet read a `show` command. That is deliberate: this first Break & Fix is about *placing* a fault, not fixing one, because deciding which layer to investigate is the decision that all the later chapters depend on.

Three reports arrive within an hour.

Three reports, three different layers, and no commands yet.

What would you check first — and what do you expect to see?

CHECKPOINT 1 of 3

A user reports that a web page will not load. Ping to the server's IP address succeeds. Which layers have you just ruled out, and which is the first one you should now suspect?

Discuss · then advance

Checkpoint 1 of 3

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Layers 1 to 3 are ruled out: something reached the server and came back. Suspect layer 7 first — specifically name resolution or the web service itself. Test the name (Chapter 21) and the port (Chapter 6).

CHECKPOINT 2 of 3

A switch port shows as up, but the collisions counter is increasing steadily. Which layer is at fault, and what is the most likely single cause?

Discuss · then advance

Checkpoint 2 of 3

A switch port shows as up, but the collisions counter is increasing steadily. Which layer is at fault, and what is the most likely single cause?

Layer 1. Rising collisions on a link that reports full duplex means a duplex mismatch: one end is half duplex and is correctly reporting collisions the other end does not see.

CHECKPOINT 3 of 3

Why does a router drop a packet for an unknown network when a switch floods a frame for an unknown MAC address?

Discuss · then advance

Checkpoint 3 of 3

Why does a router drop a packet for an unknown network when a switch floods a frame for an unknown MAC address?

Because they are different jobs. A switch's flooding is a *recovery* mechanism — the destination is probably on one of those ports and will answer, teaching the switch where it is. A router's unknown network has no such prospect; flooding a packet to every interface would loop and multiply. So a router drops and reports.

What to take away

- Networks solve four problems: addressing, forwarding, media access and error handling. Every protocol you meet answers one of them.
- Layers matter because they tell you what to test next. Work upwards from the physical layer; do not start at DNS.
- Switches forward frames using MAC addresses and flood when they do not know. Routers forward packets using IP addresses and drop when they do not know.
- One VLAN is one broadcast domain. Every switch port is its own collision domain.
- CCNA v2.0 has twenty-nine topics. Sixteen of them ask you to configure or to fix. Study accordingly.

Where we got to

- Networks solve four problems: addressing, forwarding, media access and error handling. Every protocol you meet answers one of them.
- Layers matter because they tell you what to test next. Work upwards from the physical layer; do not start at DNS.
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NEXT

Chapter 2 — The Physical Layer and Cable Faults